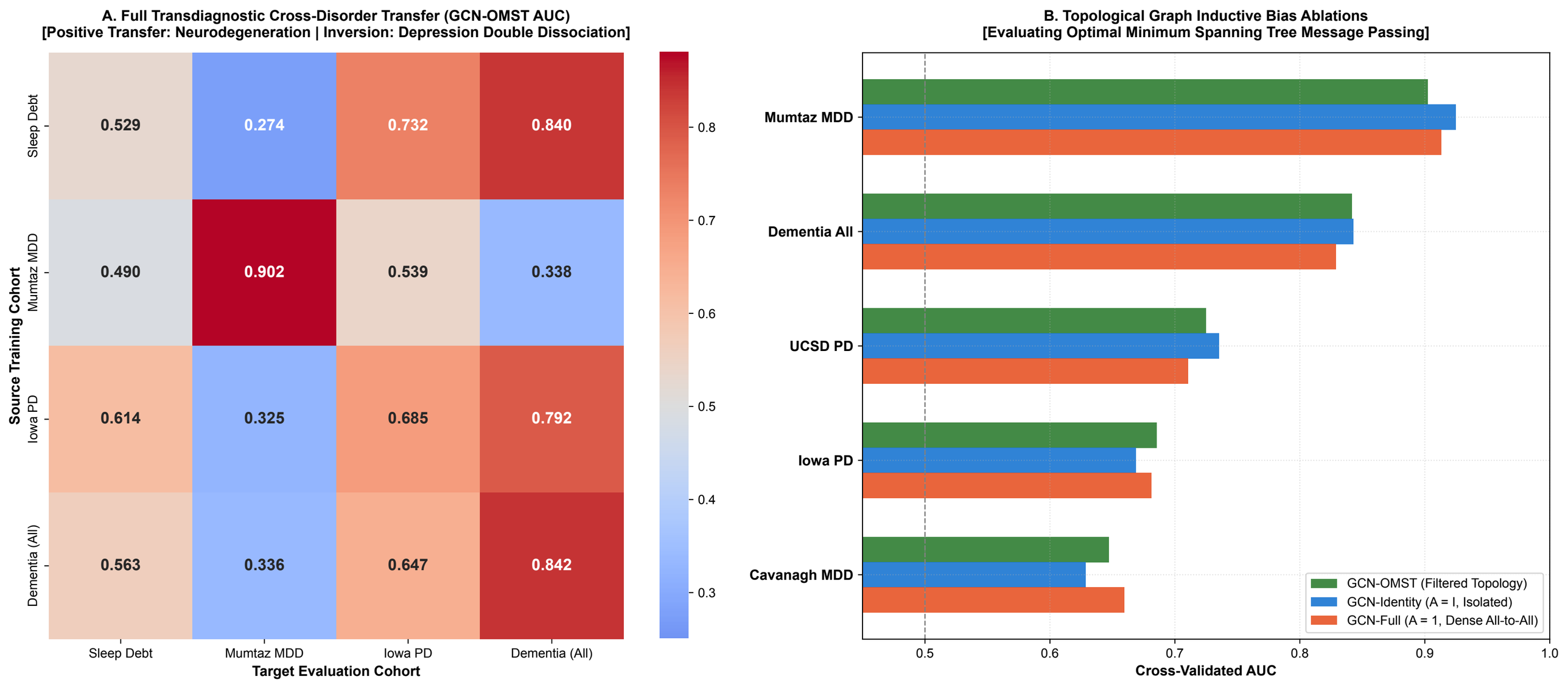
A. Full Transdiagnostic Cross-Disorder Transfer (GCN-OMST AUC)
[Positive Transfer: Neurodegeneration | Inversion: Depression Double Dissociation]
Source Training Cohort
Sleep Debt
Mumtaz MDD
Iowa PD
Dementia (All)
0.529
0.274
0.732
0.840
0.490
0.902
0.539
0.338
0.614
0.325
0.685
0.792
0.563
0.336
0.647
0.842
Sleep Debt
Mumtaz MDD
Iowa PD
Dementia (All)
Target Evaluation Cohort
0.8
0.7
0.6
0.5
0.4
0.3
B. Topological Graph Inductive Bias Ablations
[Evaluating Optimal Minimum Spanning Tree Message Passing]
Mumtaz MDD
Dementia All
UCSD PD
Iowa PD
Cavanagh MDD
GCN-OMST (Filtered Topology)
GCN-Identity (A = I, Isolated)
GCN-Full (A = 1, Dense All-to-All)
0.5
0.6
0.7
0.8
0.9
1.0
Cross-Validated AUC